Complaint Management System v.1.0
Project Overview
This project is a web-based complaint management system developed using PHP and MySQL. The system allows users to register accounts, log in securely, submit complaints, and manage complaint records through a structured database system.

The project was developed as a practical learning environment to strengthen understanding in:

- PHP fundamentals
- MySQL database integration
- CRUD operations
- User authentication
- Session management
- Basic web application structure

Objectives
The main objectives of this project are:

- Build a functional web-based complaint system
- Implement secure user registration and login
- Establish PHP to MySQL database connectivity
- Apply CRUD operations in real scenarios
- Implement session-based authentication
- Simulate a structured backend workflow

Technologies Used
- PHP
- MySQL Community Server
- MySQL Workbench
- Apache (XAMPP)
- HTML5
- CSS3
- Visual Studio Code

System Features
- User Authentication
- User registration
- Secure password hashing
- Login verification using password_verify()
- Session-based access control
- Logout functionality
- Complaint Management

- Submit complaint records
- Store complaint data into MySQL database
- Display complaint history
- User-specific complaint isolation
- Status tracking system
- Database Management
- Relational database structure
- Primary key and foreign key implementation
- One-to-many relationship between users and complaints
- JOIN query implementation

Database Structure
The system consists of two main tables:
users

Stores:
- User information
- Email credentials
- Password hash
- Registration timestamp
- aduan
- Stores:
- Complaint details
- Complaint status
- Submission timestamp
- User relationship via foreign key

Relationship:
users.id → aduan.user_id
One user can submit multiple complaints.

Security Implementation
Basic security practices implemented in this project include:
- Password hashing using password_hash()
- Password verification using password_verify()
- Session-based authentication
- Restricted page access
- Redirect control after login and form submission
- Duplicate email prevention

Development Process
The project development was divided into several phases:

V1.1 Environment setup and database connectivity
V1.2 Database configuration and relational structure
V1.3 User registration module
V1.4 Login and session control
V1.5 Complaint management module

Testing Performed
The following tests were performed during development:

- Apache and PHP execution testing
- MySQL connection testing
- User registration testing
- Password hashing verification
- Login authentication testing
- Session access restriction testing
- Complaint submission testing
- SQL JOIN query testing

Login Page
System Features → User Authentication

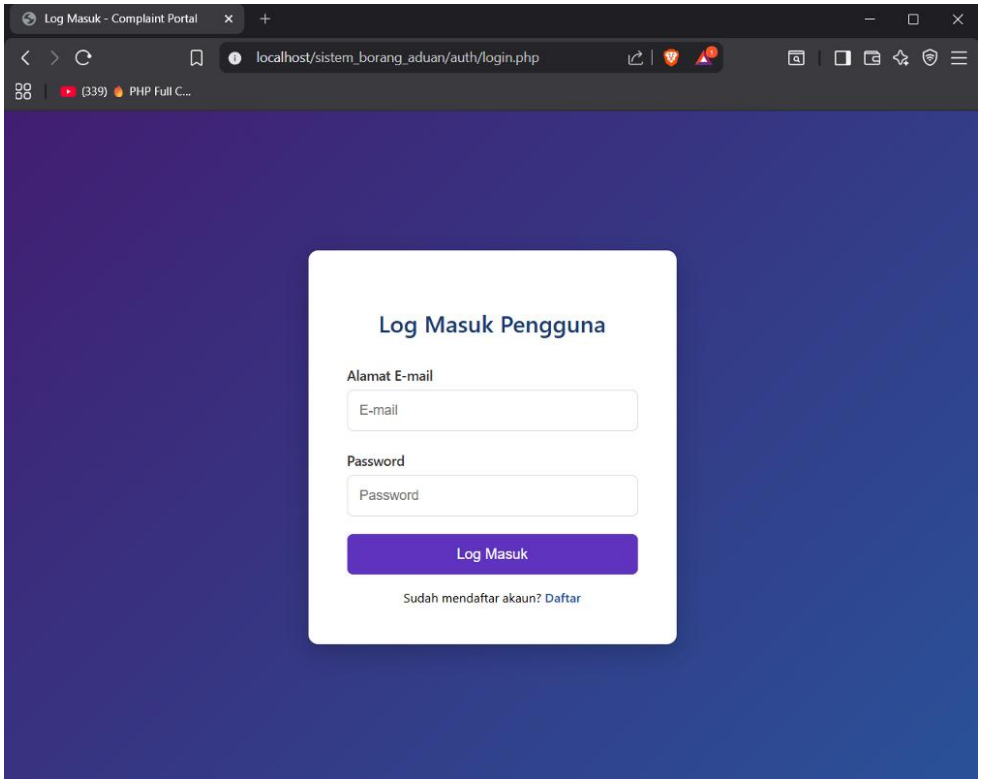
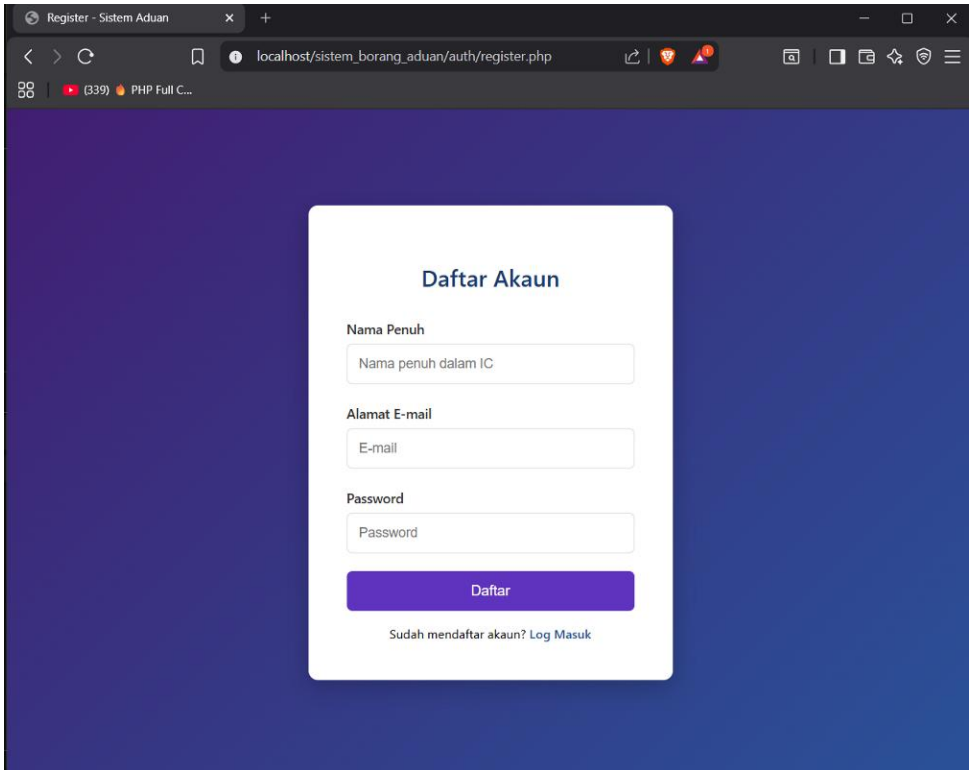


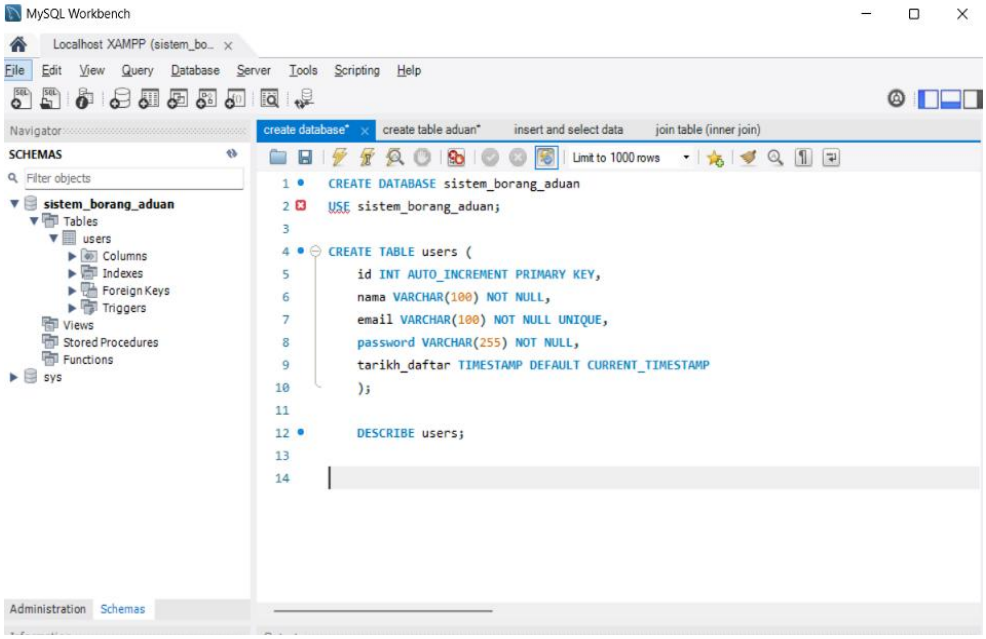
Figure 1: User Login Interface

Registration Page



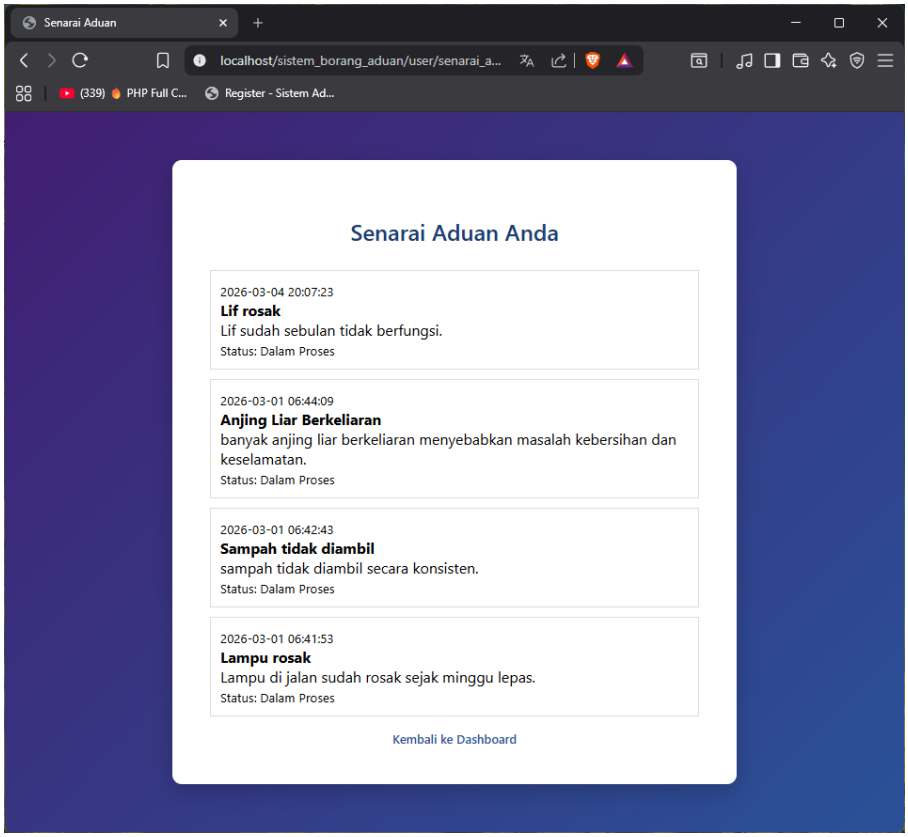
User Registration Form

Database Tables (MySQL Workbench)



Database Tables Structure in MySQL Workbench

Complaint List / Dashboard



Display complaint history

Final Section

9. Conclusion

This project successfully demonstrates the implementation of a basic web-based complaint management system using PHP and MySQL. The development process strengthened practical understanding in backend development, database integration, authentication handling, and structured system design.

The project also provided hands-on experience in troubleshooting, database configuration, and real-world application workflow simulation.